

UCS414: COMPUTER NETWORKS

L	T	P	Cr
3	0	2	4.0

Course Objectives: The subject will introduce the basics of computer networks to students through a study of layered models of computer networks and applications.

Introduction: Computer Network, Criteria and Energy Efficient Networking, Green IT Practices, Classification of networks, Network performance and Transmission Impairments. Networking Devices, OSI and TCP/IP Protocol Suite, Layering principles, Energy efficient Line Encoding, Switching and Multiplexing techniques.

Local Area Networks: Networking topologies: Bus, Star, Ring, Token passing rings, Ethernet, Environmental Considerations in Network Design, Energy Optimization in LAN, IEEE standards 802.3, 802.4, 802.5.802.11. Multiple access: ALOHA, Slotted ALOHA, Hidden and Exposed Terminal problems, Carrier sense multiple access protocols, CSMA/CD, CSMA/CA, Power Optimized Wireless LANs.

Reliable Data Delivery: Error Detection, Parity Check, Checksum and CRC, Error control (Sustainable retransmission techniques, timers), Energy Optimized Flow control (Acknowledgements, sliding window), Multiple Access, Performance issues (pipelining).

Routing and Forwarding: Routing versus forwarding, Static and dynamic routing, Unicast and Multicast Routing. Distance-Vector, Link-State, Shortest path computation, Dijkstra's algorithm, Network Layer Protocols (IP, ICMP), IP addressing, Sub-netting, IPV6, Address binding with ARP, Carbon/Energy-aware Routing Strategies, and optimization techniques for reducing the carbon footprint of network traffic (Green Data Centres and Servers).

Process-to-Process Delivery: UDP, TCP and SCTP, Multiplexing with TCP and UDP, Principles of congestion control, Sustainable Approaches to Congestion control, Balancing Quality of Service and Energy, Flow characteristics, Techniques to improve QoS.

Sustainable Network Applications: Naming and DNS, Uniform Resource Identifiers, Energy Efficient Distributed Applications (client/server, peer-to-peer, Smart and Sustainable Cities, etc.), File transfer, Telnet, e-mail, Bluetooth.

Laboratory work:

To design conceptual networks using tools like E-Draw, Microsoft Visio, and NS-3, implement topologies (BUS, RING, STAR, and Mesh) in GNS3, and develop skills in configuring IP Addresses, Routers, DHCP, and Sub-netting.

Course Learning Outcomes (CLOs) / Course Objectives (COs):

After the completion of the course, the student will be able to:

1. Conceptualize and explain the functionality of the different layers within a network architecture
2. Understand the concept of data communication, error detection and correction, access and flow control.
3. Demonstrate the operation of various routing protocols, sub-netting and their performance analysis.
4. Illustrate design and implementation of datalink, transport and network layer protocols within a simulated/real networking environment.

Text Books:

1. Forouzan A. B., Data communication and Networking, McGraw Hill (2012) 5th ed.
2. Tanenbaum S. A. and Wetherall J. D., Computer Networks, Prentice Hall (2013) 5th ed.

Reference Books:

1. Kurose J. and Ross K., Computer Networking: A Top Down Approach, Pearson (2017) 7th ed.
2. Stallings W., Computer Networking with Internet Protocols and Technology, Pearson (2004).